

The Critical Core Radius: A Geometric Certificate for Phone Separability in Frame-Level MFCC Space

Abstract

Density-based clustering fails on frame-level MFCC features, and the failure is usually reported without attribution: the same negative result is produced by a poor algorithm choice, by poor hyperparameters, and by classes that genuinely overlap. We separate these causes with a controlled ablation. For each phone we fit a full-covariance Gaussian and draw Metropolis–Hastings samples restricted to a Mahalanobis ball of radius τ around the class mean, which preserves the class means and covariance orientations while making the amount of tail mass a free parameter. Clustering quality then becomes a curve in τ rather than a single verdict. We show that this curve is monotone — every tightening step improves NMI — but that reading it naively is a trap: as $\tau \rightarrow 0$ the sampler’s acceptance rate collapses to zero, the chain freezes at the class mean, and DBSCAN attains NMI = 1.0 by clustering 39 point masses. We therefore introduce a quantity that requires no clustering algorithm at all. For a pair of classes, $\tau_{ij} = \min_x \max(d_M(x, \mu_i; \Sigma_i), d_M(x, \mu_j; \Sigma_j))$ is a convex program whose value is the largest radius at which the two cores are provably disjoint. Its minimum over pairs is a *certificate*: below it, separability is guaranteed by geometry. For the four acoustically distant classes **aa**, **n**, **sh**, **sil** the certificate is 1.205σ and sits in a healthy sampling regime, so the near-perfect clustering reported there is real. For all 39 phones the certificate is 0.222σ — inside the frozen regime, where acceptance is ≤ 0.002 . **No radius exists at which the 39 phone cores are simultaneously provably separated and non-degenerate.** We validate the certificate against an independent classifier: over all 741 phone pairs, τ_{ij} predicts held-out confusion with Spearman $\rho = -0.849$ ($p \approx 10^{-206}$), and 14 of the 20 tightest pairs are among the 20 most confused. Per phone, accuracy is predicted not by the nearest competitor ($\rho = 0.13$, n.s.) but by *crowding*, the mean of the ten smallest τ_{ij} ($\rho = 0.80$, $p = 7 \times 10^{-10}$). Finally we release a 39-way K -shot support set of Mahalanobis-ranked real frames with honest held-out baselines: 39-way top-1 accuracy is 0.368 against a chance rate of 0.026.

1 Introduction

Clustering frame-level acoustic features is the most direct formulation of unsupervised phone-unit discovery, and density-based methods such as DBSCAN are a natural fit: they do not require the number of units in advance, and they encode the intuition that a phone is a dense region of acoustic space. On MFCC frames they do not work. Tuned aggressively they either label the majority of frames as noise or collapse everything into one cluster.

That much is known. The difficulty is attribution. A negative clustering result is compatible with at least three explanations — the wrong algorithm, the wrong hyperparameters, or a representation in which the classes are not separable at all — and reporting the failure does not distinguish them. This paper is about distinguishing them.

Our instrument is a synthetic dataset that inherits its class geometry from the real data but exposes the overlap as a knob. We fit one full-covariance Gaussian $\mathcal{N}(\mu_k, \Sigma_k)$ per phone and sample from it truncated to $\{x : d_M(x, \mu_k) \leq \tau\}$. At large τ the sample reproduces the overlap of the real data; as τ shrinks the classes contract toward their means and the overlap is removed in a

controlled way. Running an identical clustering sweep at each τ converts a binary failure into a curve.

The curve alone, however, is not enough, and the reason is the main methodological point of this paper. As $\tau \rightarrow 0$ every class contracts to a point and any algorithm trivially succeeds, so a sufficiently small τ always “works”. We make this concrete rather than treating it as a caveat: the Metropolis–Hastings acceptance rate is a direct measure of whether the core is still a distribution, and at $\tau \leq 0.25\sigma$ with 39 classes it falls to 0.002 or below. The apparently perfect clustering in that regime is an artifact of a frozen chain.

This motivates a quantity defined without reference to any clustering algorithm. For two classes we ask for the smallest radius at which their cores touch,

$$\tau_{ij} = \min_x \max (d_M(x, \mu_i; \Sigma_i), d_M(x, \mu_j; \Sigma_j)),$$

a convex program with a unique optimum. If $\tau < \tau_{ij}$ the two truncated supports are disjoint sets, so the minimum of τ_{ij} over pairs is a sufficient condition for separability that no algorithm can improve on. Our contributions are:

1. **A tail-truncation ablation** attributing the failure of DBSCAN on MFCC frames to class overlap rather than to the algorithm, and turning it into a monotone curve in τ (§6).
2. **An explicit degeneracy diagnostic.** The MH acceptance rate identifies the regime in which apparent clustering success is an artifact, which we show is exactly where the 39-class problem becomes separable (§6).
3. **The certificate** τ_{ij} (§7), computed for all 741 phone pairs, together with its scaling in inventory size: the certificate falls from 1.64σ at $K = 2$ to 0.222σ at $K = 39$.
4. **External validation** (§8): τ_{ij} predicts the held-out confusions of an unrelated classifier at $\rho = -0.849$, and a crowding statistic derived from it predicts per-phone accuracy at $\rho = 0.80$.
5. **A released support set and honest baselines** (§9), including a correction to previously reported episode accuracies that were measured with queries drawn from the support pool.

2 Related Work

Density-based clustering. DBSCAN [1] defines clusters as regions of high density separated by regions of low density, with a global radius ε and a minimum neighbourhood count. Its known weakness is data with variable density, which motivated OPTICS [2] and HDBSCAN [3]; we evaluate all three.

Acoustic unit discovery. Unsupervised discovery of subword units has been driven by the ZeroSpeech challenge series [4], with Dirichlet-process mixture approaches [5] among the strongest frame-level clustering baselines. Our conclusion — that the limiting factor is the representation — is consistent with the field’s move toward learned features.

Learned speech representations. wav2vec 2.0 [6], HuBERT [7] and WavLM [8] produce representations whose intermediate layers are substantially more phonetically linear than MFCCs. Recomputing our certificate on those layers is the natural continuation (§10).

Few-shot classification. Matching Networks [9] and Prototypical Networks [10] learn embeddings in which nearest-neighbour or nearest-centroid classification succeeds from a handful of examples. The support set we release is in exactly the form these methods consume.

3 Data and Setup

We use the TIMIT [11] training partition: 4,620 utterances, each stored as a $(T, 12)$ matrix of MFCC frames with a frame-level phone transcript of (label, start, end) triples in frame units. Expanding the transcripts gives 1,419,678 labelled frames over the standard 39-symbol reduced inventory [12], including `sil`.

The frame distribution is severely skewed: `sil` alone accounts for 367,150 frames (25.9%) and the smallest class `b` for 3,823. All experiments therefore use a class-stratified subset — 1,500 frames per phone at $K = 39$, 3,000 at $K = 4$ — sampled without replacement. This is not a convenience: a density method estimates one global density scale, and on unstratified counts that scale is set by silence alone.

Features are standardized with a `StandardScaler` fit on the stratified subset. Standardization is necessary because c_0 (log-energy) has a numeric range roughly an order of magnitude larger than c_1 – c_{11} , which carry most of the phonetic contrast, while DBSCAN scores neighbourhoods with an unweighted Euclidean metric. All runs use seed 0.

4 The Failure on Real Frames

We first establish that the failure is a property of the data rather than of one algorithm. On a five-class subset (`s`, `ih`, `aa`, `iy`, `n`; `sil` excluded because it dominates)¹ on a balanced 10,000-frame sample, we swept DBSCAN over $\varepsilon \in [1.6, 2.4]$ and `min_samples` $\in \{5, 8, 10, 15, 20\}$ (45 configurations, with the ε grid set from k -distance quantiles), HDBSCAN over `min_cluster_size` $\in [30, 1000]$, OPTICS over $\xi \in [0.03, 0.30]$, a 132-dimensional context-window variant (± 5 frames, PCA-whitened to 30 components retaining 95% variance), and a 5-component full-covariance GMM as a non-density reference.

Table 1: Five-class comparison on 10,000 balanced frames. The best density-based NMI covers 17% of the data; the best NMI overall comes from a model that is not density-based at all.

Method	Parameters	Clusters	Coverage	NMI	ARI
DBSCAN	$\varepsilon = 1.6$, $ms = 8$	5	0.633	0.109	0.078
DBSCAN	$\varepsilon = 1.4$, $ms = 8$	11	0.351	0.163	0.108
HDBSCAN	$mcs = 50$	2	0.166	0.233	0.069
DBSCAN (132D ctx)	$\varepsilon = 2.4$, $ms = 8$	5	0.127	0.225	0.064
OPTICS	$ms = 10$, ξ sweep	0	0.000	0.000	0.000
GMM	5 full-cov. components	5	1.000	0.555	0.525

Three readings set up the rest of the paper. The density-based family trades coverage against purity and never obtains both: the context-window variant reaches 0.978 clustered purity on 12.7% of frames, and HDBSCAN’s honest answer is two clusters, one an almost perfectly pure `s`. OPTICS finds no usable reachability valleys at all, indicating that the density transitions are flat rather than merely mis-thresholded — visible directly in the k -distance curves (Figure 1, right), which rise smoothly with no elbow. And a GMM, which models each class as an overlapping ellipsoid instead

¹An earlier internal report describes this subset as the five densest non-silence classes, which would substitute `ae` for `n`. The run that produced Table 1 used `n`: its pool of 352,543 frames matches $84,491 + 84,270 + 73,885 + 62,901 + 46,996$ exactly.

of thresholding a global density, roughly doubles the best density-based NMI at full coverage. All three point at overlapping class supports.



Figure 1: Left: t-SNE of 4,000 raw standardized frames from the four-class subset, coloured by true phone. **sh** forms a distinct cloud; **aa**, **n** and **sil** overlap along their boundaries. Right: sorted distance to the k -th nearest neighbour at $k = 8, 12, 20$. The absence of a sharp elbow is why no single ε works.

5 Method: Core Sampling

5.1 Per-class Gaussians

At $K = 4$ we fit a four-component full-covariance GMM by EM with `means_init` set to the per-phone sample means and `weights_init` to the per-phone frequencies, then reorder components by nearest mean. The recovered mapping is `{aa, n, sh, sil}` with weights `[0.256, 0.417, 0.247, 0.081]`, i.e. EM does keep each component bound to its intended phone at this scale.

At $K = 39$ this is no longer safe: joint EM over 39 components re-mixes them and destroys the component-to-phone binding on which the whole diagnostic depends. We therefore fit each Gaussian independently on its own class-conditioned frames,

$$\mu_k = \frac{1}{N_k} \sum_{i:y_i=k} x_i, \quad \Sigma_k = \frac{1}{N_k - 1} \sum_{i:y_i=k} (x_i - \mu_k)(x_i - \mu_k)^\top + \lambda I,$$

with $\lambda = 10^{-4}$ as a numerical guard, which makes the binding hold by construction.

5.2 Truncated sampling

For component k we sample from

$$\pi_k(x) \propto \mathcal{N}(x; \mu_k, \Sigma_k) \mathbf{1}[d_M(x, \mu_k) \leq \tau], \quad d_M(x, \mu_k) = \sqrt{(x - \mu_k)^\top \Sigma_k^{-1} (x - \mu_k)},$$

by Metropolis–Hastings with a symmetric Gaussian random-walk proposal $x' = x + s \Sigma_k^{1/2} \eta$, $\eta \sim \mathcal{N}(0, I)$. Symmetry reduces the acceptance ratio to

$$\alpha(x \rightarrow x') = \min\left(1, \exp\left[-\frac{1}{2}(d_M^2(x', \mu_k) - d_M^2(x, \mu_k))\right]\right) \mathbf{1}[d_M(x', \mu_k) \leq \tau].$$

We use burn-in 500, thinning 3, and $s = 0.25$ for $\tau \geq 1.2$, $s = 0.15$ below. The indicator is the entire point of the construction: it deletes the tails, where adjacent phones meet, and leaves the means and covariance orientations untouched.

5.3 Selection rule

At each τ we sweep $\varepsilon \in [0.10, 1.00]$ in steps of 0.05 and $\text{min_samples} \in \{5, 10\}$, and select the configuration maximizing NMI subject to a noise budget of 0.4. The noise budget is not optional. Ranking by proximity to the target cluster count instead rewards a degenerate solution: at $\tau = 1.5$, $K = 39$ that rule selects a run with 38 clusters of which one holds 4,860 points spanning 21 phones at 10% purity, while 71% of points are noise.

6 The τ Sweep and the Degenerate Regime

Table 2: DBSCAN on MCMC cores as a function of core radius τ , 250 samples per class, best configuration per row under the noise budget. NMI improves monotonically as τ shrinks — but so does the acceptance rate, and the top rows are sampling a frozen chain rather than a distribution.

τ	MH accept.	ε	ms	Clusters	Noise	NMI	ARI
0.15	0.000	0.15	5	39	0.000	1.000	1.000
0.20	0.000	0.25	5	39	0.000	1.000	1.000
0.25	0.002	0.35	5	39	0.000	1.000	1.000
0.30	0.006	0.35	10	44	0.001	0.985	0.951
0.40	0.028	0.45	5	34	0.000	0.973	0.862
0.50	0.075	0.40	5	52	0.007	0.951	0.825
0.60	0.134	0.45	5	50	0.009	0.924	0.741
0.75	0.232	0.50	5	54	0.026	0.780	0.325
1.00	0.370	0.55	5	249	0.132	0.644	0.167
1.25	0.227	0.65	5	368	0.258	0.551	0.090
1.50	0.309	0.75	5	343	0.238	0.459	0.048

The monotone trend is unambiguous: every tightening of τ improves NMI, from 0.459 at $\tau = 1.5$ to 1.000 at $\tau \leq 0.25$. Read alone, the table says the classes are recoverable. The acceptance column says otherwise. At $\tau \leq 0.25$ the Metropolis chain accepts 0.2% of proposals or fewer: it never leaves its initial state at μ_k , so the “cluster” for each phone is 250 copies of the class mean, and DBSCAN is separating 39 point masses. The perfect scores in the top three rows measure nothing about the representation.

The same protocol on the four-class subset behaves completely differently. There, $\text{NMI} = 1.000$ holds across $\tau \in [0.25, 1.75]$ with acceptance between 0.08 and 0.39 — a wide window in which the cores are genuine distributions *and* DBSCAN recovers them. This contrast, shown in Figure 2, is the substantive result: the four-class success is real and the 39-class success is not.

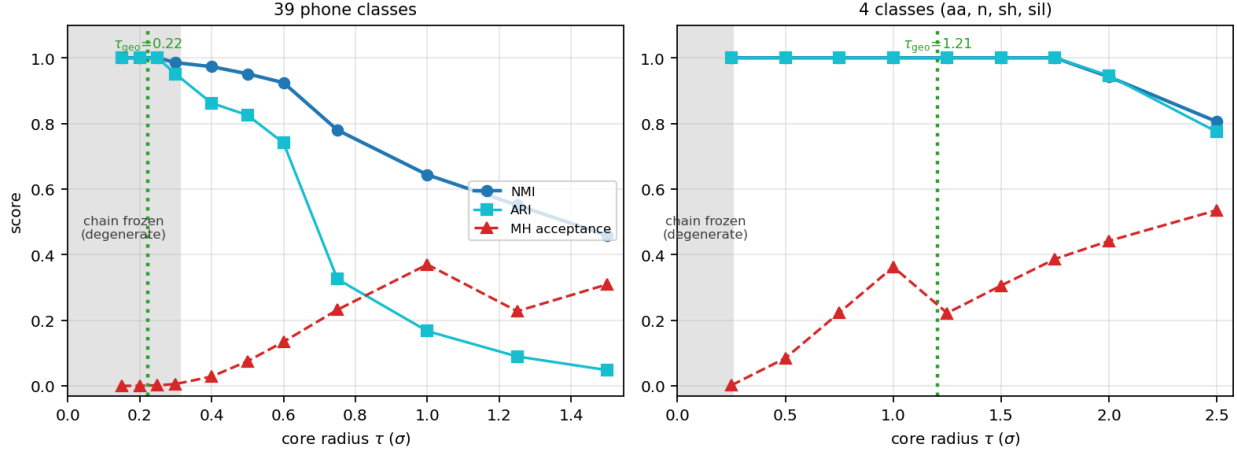


Figure 2: Clustering quality and MH acceptance rate against core radius τ . Shaded: the regime where acceptance < 0.02 and the sampler has frozen at the class mean. Dotted green: the geometric certificate τ_{geo} of §7. *Left*, 39 classes — the certificate lies inside the frozen region, so there is no radius that is both certified and non-degenerate. *Right*, four classes — the certificate lies well outside it, and a wide valid window exists.

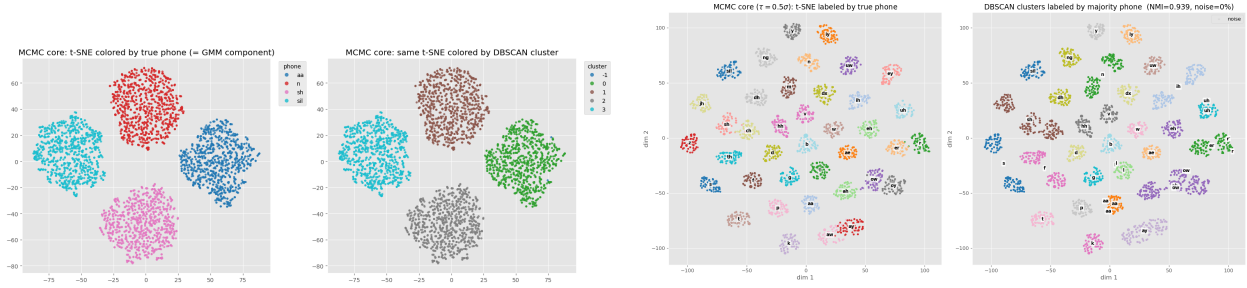


Figure 3: Left: four-class cores at $\tau = 1.5$, truth versus DBSCAN — the panels are visually identical. Right: 39-class cores at $\tau = 0.5$, with each cluster annotated by its majority phone. The cluster count is close to target but the map is not bijective: adjacent blobs share cluster identities.

Relation to a previously reported result. An earlier run of this experiment (400–500 samples per class, coarser ε grid, ranking that also rewarded proximity to 39 clusters) reported exactly 39 clusters at $\tau = 0.5$ with $\text{NMI} = 0.939$ and noise 0.0018, and was summarized as a recovery of the inventory. The re-run in Table 2 reaches a similar NMI (0.951) at the same τ but finds 52 clusters, and the earlier run’s own composition analysis had already shown that only 31 of 39 phones were the majority of any cluster, with 8 phones (ih, ay, z, sh, m, oy, jh, th) absorbed into neighbours. The exact cluster count is therefore sensitive to the sweep grid and to the selection rule, and should not be read as recovery of the inventory. The monotone τ trend is robust across both runs; the cluster count is not.

7 A Geometric Certificate

The acceptance-rate diagnostic tells us when a τ is degenerate, but it is still tied to a sampler and a clustering algorithm. We now remove both.

7.1 Definition

For classes i and j define

$$\tau_{ij} = \min_{x \in \mathbb{R}^{12}} \max(d_M(x, \mu_i; \Sigma_i), d_M(x, \mu_j; \Sigma_j)). \quad (1)$$

Each $d_M(\cdot, \mu_k; \Sigma_k)$ is a norm of an affine function of x and therefore convex; the pointwise maximum of convex functions is convex; so (1) is a convex program with a unique optimal value, solved here in its epigraph form $\min t$ s.t. $d_M(x, \mu_i) \leq t$, $d_M(x, \mu_j) \leq t$ by SLSQP with analytic gradients.

The interpretation is immediate. The truncated supports $B_i(\tau) = \{x : d_M(x, \mu_i) \leq \tau\}$ and $B_j(\tau)$ intersect if and only if there exists x with both distances $\leq \tau$, i.e. iff $\tau \geq \tau_{ij}$. Hence for $\tau < \tau_{ij}$ the two cores are disjoint sets, and

$$\tau_{\text{geo}} = \min_{i \neq j} \tau_{ij}$$

is a radius below which *all* class cores are pairwise disjoint. This is a statement about the class geometry alone: no clustering algorithm, no sampler, no hyperparameter.

Note the direction of the guarantee. $\tau < \tau_{\text{geo}}$ is sufficient for disjoint supports, not necessary for a clustering method to succeed — with finite samples the region near the touching point is very sparsely populated, so DBSCAN can and does succeed at τ somewhat above τ_{geo} . The certificate is a conservative bound, and both empirical windows respect it.

7.2 Results

Table 3: The ten least-separable phone pairs by certificate, with the Euclidean distance between class means for comparison. Phones marked † are those absorbed by DBSCAN in the 39-class run.

Pair	τ_{ij}	$\ \mu_i - \mu_j\ $	Pair	τ_{ij}	$\ \mu_i - \mu_j\ $
ay†/aw	0.222	0.496	ch/jh†	0.416	0.595
sh†/ch	0.258	0.348	ih†/ey	0.442	0.874
er/r	0.307	0.514	ah/ow	0.445	0.983
ow/oy†	0.322	0.479	n/m†	0.479	0.764
sh†/jh†	0.379	0.516	aa/aw	0.486	1.096

For the full inventory $\tau_{\text{geo}} = 0.222\sigma$, set by ay/aw. For the four-class subset $\tau_{\text{geo}} = 1.205\sigma$, set by n/sil. Placing these on Figure 2 gives the paper’s central claim:

At $K = 39$ the certificate (0.222σ) lies inside the frozen regime (acceptance ≤ 0.002). Certified separability and non-degenerate sampling have no common radius. At $K = 4$ the certificate (1.205σ) lies in a healthy regime (acceptance 0.36 at $\tau = 1.0$), and the observed recovery is genuine.

The certificate is correlated with the distance between class means ($r = 0.887$) but is not a rescaling of it, and the disagreements are informative. By mean distance the hardest pair is sh/ch (0.348); by certificate it is ay/aw, whose means are 42% further apart (0.496) but whose covariances are oriented so that the ellipsoids reach toward each other. Diphthongs are exactly the classes for which a frame-level Gaussian is widest, and the certificate accounts for that where a mean distance cannot.

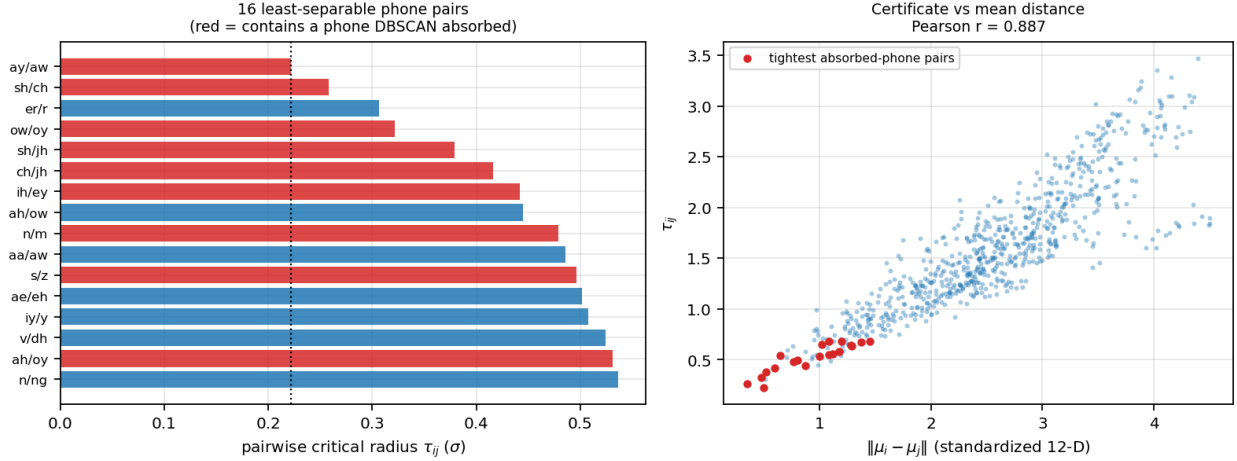


Figure 4: Left: the 16 least-separable pairs; red bars contain a phone that DBSCAN absorbed. Right: certificate against distance between class means over all 741 pairs.

7.3 Scaling with inventory size

Because τ_{ij} is defined per pair, τ_{geo} for any subset of classes is a minimum over a submatrix, and the scaling of separability with inventory size can be computed exactly without re-running anything. Averaging over 400 random subsets per size:

Table 4: τ_{geo} against inventory size K , mean $\pm$ sd over random subsets.

K	2	4	8	12	16	24	32	39
τ_{geo}	$1.64 \pm .60$	$0.88 \pm .27$	$0.54 \pm .16$	$0.42 \pm .13$	$0.35 \pm .10$	$0.29 \pm .07$	$0.24 \pm .03$	0.222

Separability decays steeply and then saturates: by $K \approx 24$ the certificate is essentially pinned by the single worst pair present in almost every subset. The four-class set actually used (**aa**, **n**, **sh**, **sil**, $\tau_{\text{geo}} = 1.205$) sits near the 90th percentile of random four-class subsets ($p_{90} = 1.234$) — it was chosen for acoustic distinctness, and it is quantifiably an easy case. Any conclusion drawn from four hand-picked classes therefore does not transfer to the full inventory, which is precisely what the 39-class experiment found the hard way.

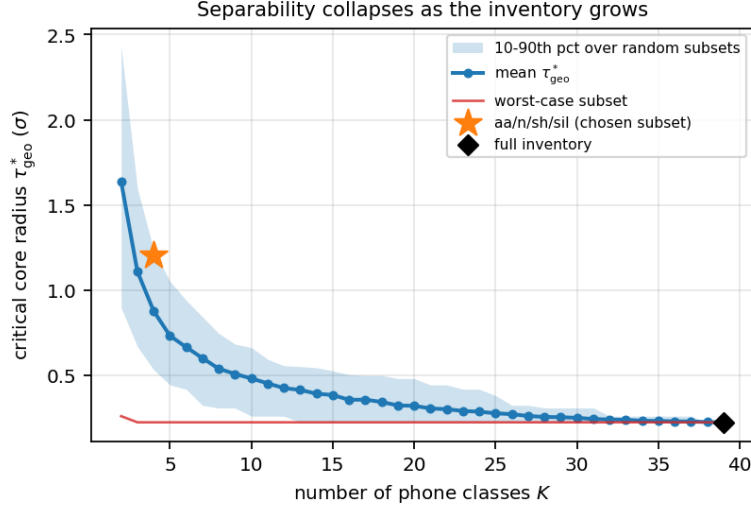


Figure 5: Certificate against inventory size. The chosen four-class subset (star) is near the top decile of its size; the full inventory (diamond) is set by **ay/aw**.

8 Validating the Certificate

A geometric quantity is only useful if it predicts behaviour that was not used to construct it. We test it against a classifier that shares no machinery with DBSCAN: quadratic discriminant analysis on *held-out real frames* (§9), for which we compute the symmetric confusion rate S_{ij} between every pair of phones.

- Over all **741 pairs**, τ_{ij} predicts S_{ij} with Spearman $\rho = -0.849$ ($p \approx 1.5 \times 10^{-206}$); restricted to the 581 pairs ever confused, $\rho = -0.803$.
- **14 of the 20** pairs with the smallest certificate are among the 20 most-confused pairs.
- The most-confused pairs are **sh/ch** (certificate rank 1 of 741), **s/z** (rank 10), **er/r** (rank 2), **iy/y** (rank 12) and **sh/jh** (rank 4).

Per-phone accuracy needs crowding, not the nearest neighbour. Our first hypothesis — that a phone’s accuracy is governed by its closest competitor, $\min_j \tau_{ij}$ — **fails**: $\rho = 0.126$, $p = 0.44$. Aggregating over the neighbourhood succeeds. The mean of a phone’s ten smallest τ_{ij} predicts its held-out accuracy at $\rho = 0.804$ ($p = 6.8 \times 10^{-10}$), and $\sum_j \tau_{ij}^{-1}$ performs comparably ($\rho = 0.802$). A phone is hard when it sits in a crowded neighbourhood, not when it has one close rival — which is consistent with a 39-way decision in which every competitor can claim probability mass.

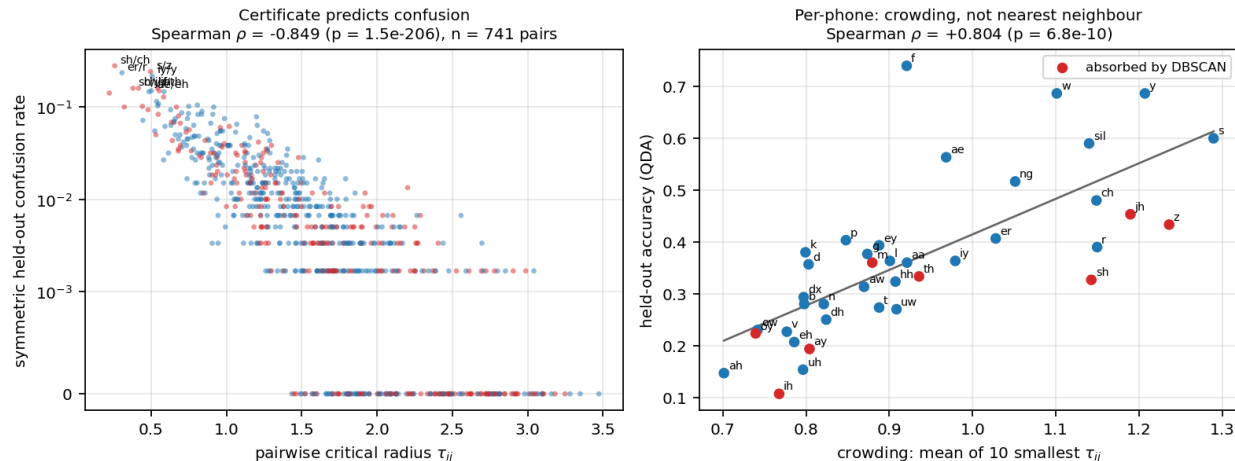


Figure 6: Left: certificate against symmetric held-out confusion for all 741 pairs (log-scaled ordinate; red points involve a DBSCAN-absorbed phone). Right: per-phone accuracy against crowding.

9 Support Set and Held-Out Baselines

The per-class Gaussians also support a constructive use. For each phone we score every real frame in the stratified subset by $d_M(x_i, \mu_k)$ and keep the $K = 10$ smallest: the most prototypical real exemplars, equivalently the K frames of highest density under the class component. This yields a 39-way 10-shot support set of 390 real frames, each traceable to its source (*utt_id, frame_idx*).

The construction uses only each phone’s own (μ_k, Σ_k) and never a pairwise comparison, so it stays well defined for the phones that clustering cannot separate; discriminating those is precisely the task handed to a downstream embedding. Per-phone prototype spread is itself interpretable: *sil* ($d_M \in [0.86, 1.08]$), *p*, *v* and *f* have the tightest cores, while *oy* ($[1.76, 2.04]$), *aw*, *r* and *ow* are widest — the diphthongs and rhotics again.

9.1 Held-out evaluation

We remove the 390 prototype frames from the corpus and sample 300 held-out frames per phone (11,700 total), standardized with the stored scaler.

Table 5: Held-out 39-way top-1 accuracy, 11,700 frames, no learned parameters.

Classifier	Accuracy
Nearest prototype mean (Euclidean)	0.277
Nearest class mean (Euclidean)	0.297
Minimum Mahalanobis to class Gaussian	0.335
Gaussian log-likelihood, full covariance (QDA)	0.368
Chance	0.026

Three points. Accuracy is $14\times$ chance yet only 0.368, which is the quantitative version of this paper’s thesis: a single 12-dimensional MFCC frame carries real but limited phone identity. The ordering matters — using the full covariance (QDA) beats using only the mean by 7 points, so the class *shape* carries information that any centroid method discards, and this is exactly the

regime where a learned metric should help. And the eight phones DBSCAN absorbed average 0.304 against 0.384 for the other 31, so the clustering diagnostic and the supervised classifier agree on which phones are hard.

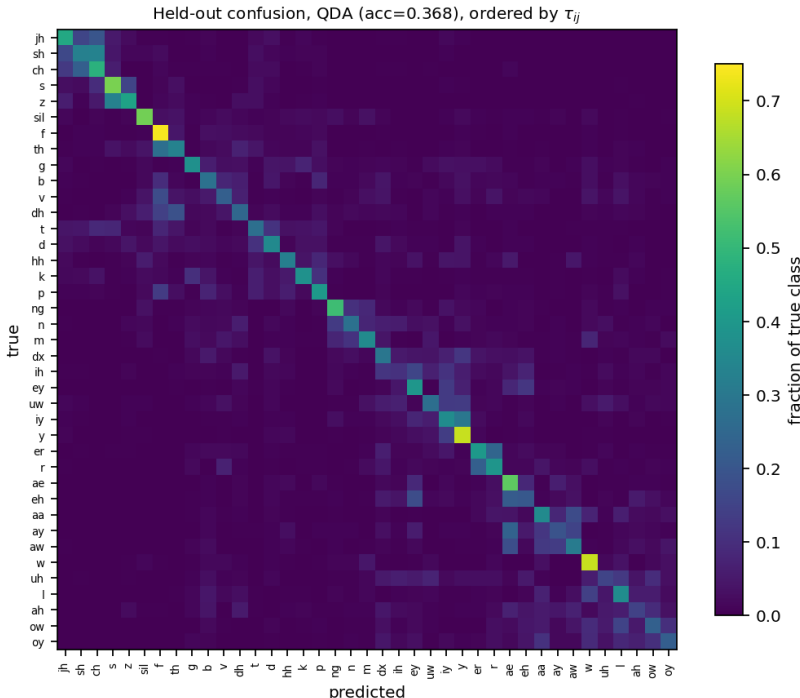


Figure 7: Held-out confusion for the QDA classifier, rows and columns ordered by hierarchical clustering of the certificate matrix. Off-diagonal mass concentrates in the blocks the certificate identifies.

9.2 A correction to previously reported few-shot numbers

Episode accuracies we reported previously drew query frames from the same pool of 390 prototypes as the supports. Since every prototype is by construction close to its class mean, this inflates accuracy substantially. Repeating the protocol with queries drawn from held-out frames:

Table 6: N -way K -shot nearest-centroid accuracy over 400 episodes. “Pool” draws queries from the prototype set (as previously reported); “held-out” draws them from unseen frames.

N -way	1-shot		5-shot	
	pool	held-out	pool	held-out
5	0.807	0.589	0.907	0.673
10	0.693	0.442	0.816	0.521
20	0.547	0.317	0.705	0.379
39	—	0.207	—	0.262

The gap is 0.20–0.23 absolute throughout. The held-out column is the number a learned few-shot model should be compared against; the pool column should not be used.

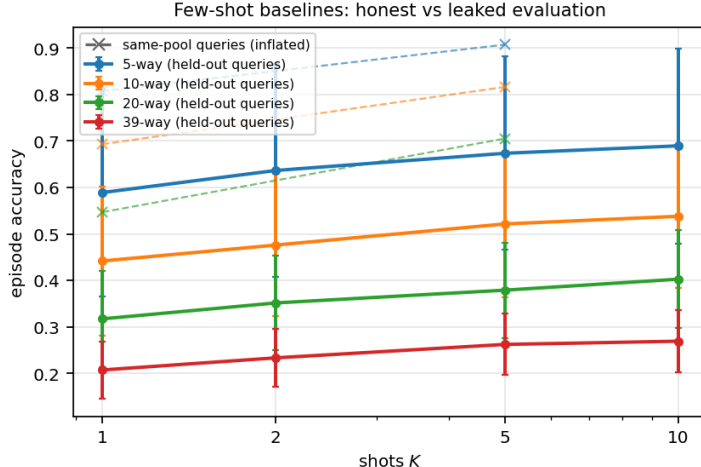


Figure 8: Few-shot baselines. Solid: held-out queries. Dashed: same-pool queries, inflated by roughly 0.2 absolute at every setting.

10 Limitations

Gaussianity. The certificate is exact for the truncated Gaussians it is defined on, and only as good as that model is for real phones. Phones with allophonic variation are plausibly multimodal, in which case a single ellipsoid overstates their extent. A per-phone mixture, with the certificate taken as a minimum over component pairs, is the natural generalization.

One feature space. Every claim here concerns 12-dimensional MFCCs. The obvious control — recomputing τ_{ij} on self-supervised representations, where we expect a far larger certificate and where the crowding statistic becomes a layer-selection criterion — is not run here and is the single most valuable next experiment.

Frame-level only. A frame is treated as i.i.d. and the trajectory structure of a phone is discarded. The one context experiment we ran (± 5 frames, 132D, PCA-whitened) raised clustered purity to 0.978 at 12.7% coverage, which is suggestive but not conclusive.

Cluster counts are grid-sensitive. As §6 documents, the exact number of clusters recovered depends on the sweep grid and selection rule; only the monotone trend and the certificate are stable.

Single corpus. TIMIT is read English, and the 39-symbol inventory already merges some contrasts.

11 Conclusion

Density-based clustering of frame-level MFCCs fails for a reason that can be measured rather than asserted. Replacing real frames with samples from per-class Gaussians truncated at a controllable Mahalanobis radius turns the failure into a monotone curve; the sampler’s acceptance rate then reveals that the regime in which the 39-class problem appears solved is one in which the cores have collapsed to points. Removing the algorithm entirely, the convex program (1) gives a certificate of separability that depends only on the class geometry: 1.205σ for four acoustically distant phones, and 0.222σ for the full inventory — the latter inside the degenerate regime, so that no valid operating point exists. The certificate is not a formal curiosity: it predicts the pairwise confusions of an unrelated held-out classifier at $\rho = -0.849$, and an aggregate of it predicts per-phone accuracy at $\rho = 0.80$. Together with a released 39-way K -shot support set and corrected baselines, this

converts a negative clustering result into a measurement instrument for the learned representations that the result says are necessary.

Reproducibility

Artifacts: `dbscan_variants_5phone.ipynb` (Table 1); `dbscan_4phone_classes_dim12.ipynb` and `dbscan_4phone_mcmc_gmm_dim12.ipynb` (four-class); `dbscan_40phone_mcmc_gmm_dim12.ipynb` (39-class, support set); `use_support_set_40phone.ipynb` (episodes); and `support_set_40phone.pkl`, which bundles the 39 class Gaussians, the scaler, the 390 prototypes with source metadata, and the DBSCAN configuration. The certificate, the re-run τ sweep, the held-out evaluation and all figures in §6–9 are produced by four standalone scripts operating on the pickle and the corpus. Seed 0 throughout.

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